SPAT0002

Coherence

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Introduction to Signal Processing

- Today's Focus:
  - Coherence
  - That’s it !

Coherence (1)

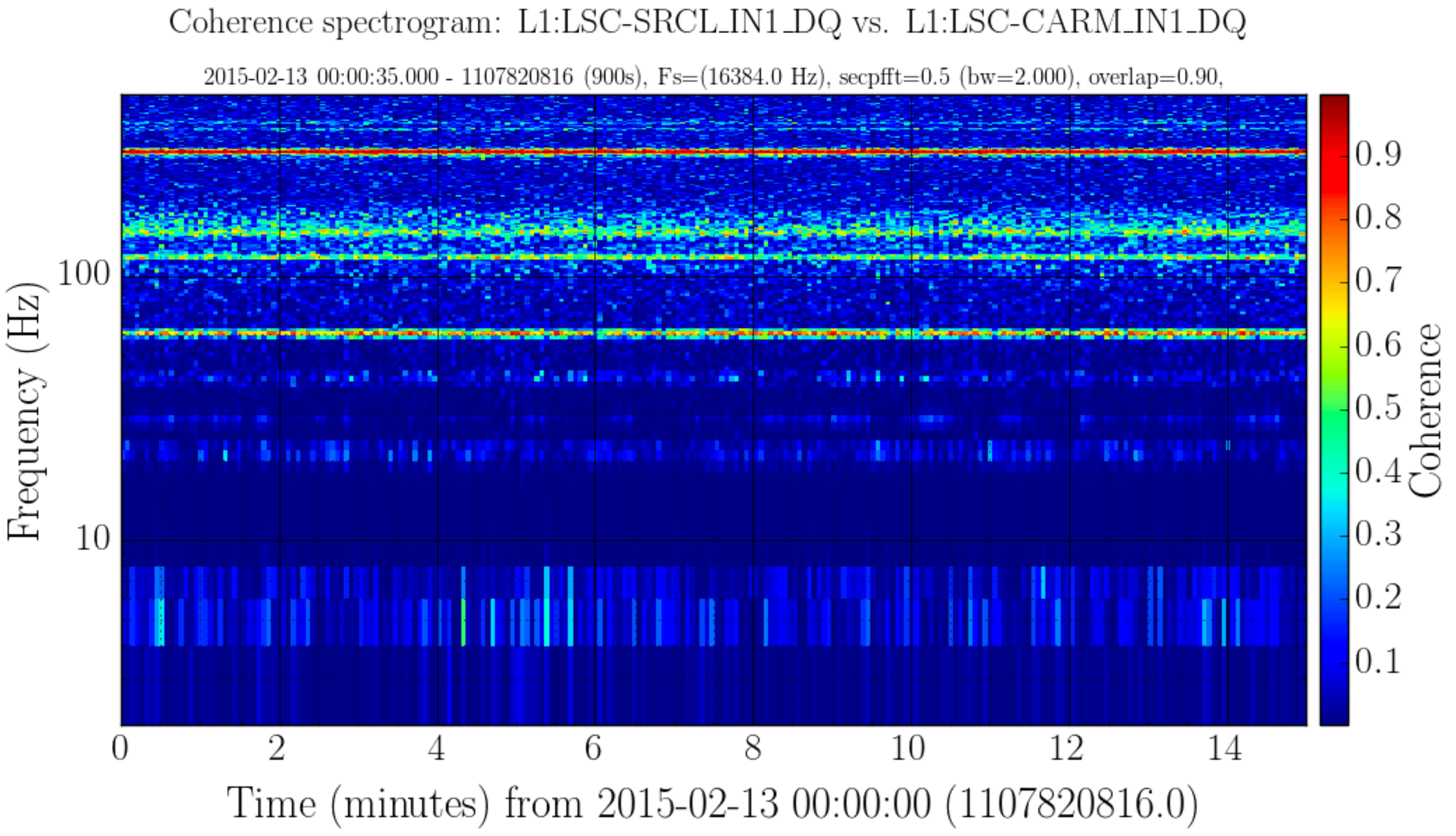
- Measures the degree of correlation between two signals at different frequencies.
- —> frequency-domain analog of the time-domain correlation coefficient.
- Used to identify the frequency components that are common in two time series.
- Essential in signal processing, neuroscience, and engineering for analyzing the synchrony of signals.
- Values range from 0 to 1.
- A value of 1 indicates perfect linear correlation at a particular frequency, while 0 indicates no correlation.
- Factors Affecting Coherence:
  - Signal-to-noise ratio (SNR) can significantly impact coherence measurements.
  - **<u>Windowing and choice of frequency resolution in spectral analysis affect coherence values.</u>**

```
Cxy = abs(Pxy)**2/(Pxx*Pyy)
```

```
Pxy = [ X(f) Y*(f) ]
```

Coherence (2)

- Need for careful interpretation in the presence of noise and non-stationarities.
- Importance of choosing appropriate window size and overlap in Fourier analysis for accurate coherence estimation.
- Applications in brain-computer interfaces, geophysics, telecommunications and gravitational wave detections.
- Challenges and Limitations:
  - Distinguishing between true coherence and artifacts due to noise or methodological issues.
  - Addressing issues of non-stationarity and non-linearity in real-world signals.


Coherence spectrogram: L1:LSC-SRCL_IN1_DQ vs. L1:LSC-CARM_IN1_DQ
2015-02-13 00:00:35.000 - 1107820816 (900s), Fs=(16384.0 Hz), secpfft=0.5 (bw=2.000), overlap=0.90,
Frequency (Hz)
100
10
0
2
4
6
8
10
12
14
Time (minutes) from 2015-02-13 00:00:00 (1107820816.0)
Coherence
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1